Skyrmion Number Modulation for Free-Space Optical Communication: A Topologically Protected Alternative to OAM and Intensity Modulation

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November 2, 2026

Abstract

We present skyrmion number modulation (SKM), a free-space optical communication scheme that encodes information in the topological skyrmion number of an optical polarization texture. Unlike orbital angular momentum (OAM) modes, which are scrambled by atmospheric turbulence, the skyrmion number is a topological invariant that remains quantized under continuous deformations. We present a complete simulation framework including atmospheric turbulence with both phase and amplitude fluctuations (scintillation), polarized field propagation, and a polarization-resolving receiver model. Under identical power, aperture, wavelength, and turbulence constraints, SKM achieves bit error rate (BER) of zero across turbulence strengths from $C_n^2 = 10^{-15}$ to 10^{-11} m$^{-2/3}$, while OAM suffers BER ≈ 0.5 and intensity modulation ties with SKM at low receiver noise but loses at high noise. In 8-state constellations, SKM achieves mutual information of $2.96/3$ bits versus $2.42/3$ bits for OAM and $1.61/3$ bits for intensity. We validate the turbulence model against von Karman spectrum theory, the Fried parameter against established formulae, and skyrmion number quantization to machine precision at all grid resolutions. All results are from mathematical simulation; physical hardware demonstration remains future work.

Keywords: skyrmion number, topological communication, free-space optics, orbital angular momentum, atmospheric turbulence, polarization modulation

1 Introduction

Free-space optical (FSO) communication offers high-bandwidth, license-free wireless links but suffers from atmospheric turbulence-induced fading and mode scrambling. Orbital angular momentum (OAM) multiplexing has been proposed as a path to high-capacity FSO links, but OAM modes are highly sensitive to turbulence-induced mode coupling [1, 2]. Intensity modulation is robust to phase distortion but requires precise power measurement and is vulnerable to scintillation.

We propose skyrmion number modulation (SKM), in which information is encoded in the topological skyrmion number Q of an optical polarization texture:

$$Q = \frac{1}{4\pi} \int \mathbf{n} \cdot (\partial_x \mathbf{n} \times \partial_y \mathbf{n})\, dx\, dy, \tag{1}$$

where $\mathbf{n} = (S_1, S_2, S_3)/S_0$ is the normalized Stokes vector. Because Q is a topological invariant, it is preserved under continuous deformations of the polarization field — including those caused by weak and moderate atmospheric turbulence — offering potential robustness advantages over both OAM and intensity modulation.

This paper makes the following contributions:

1. A complete simulation framework for polarized FSO communication through atmospheric turbulence, including phase and amplitude (scintillation) fluctuations from the von Karman spectrum.

2. Validation of the turbulence model against von Karman spectrum theory, Fried parameter formulae, and Rytov variance.

3. Validation of skyrmion number quantization to machine precision ($Q = q$ exactly for $q \in \{-4, \ldots, 4\}$) at all grid resolutions (32×32 through 512×512).

4. Fair comparison of SKM against OAM and intensity modulation under identical power, aperture, wavelength, distance, turbulence, and receiver constraints.

5. High-trial BER measurements with Clopper–Pearson 95% confidence intervals.

All results presented here are from mathematical simulation on a laptop computer. Physical hardware demonstration is required to establish experimental feasibility and is explicitly identified as future work.

2 Theory

2.1 Skyrmion number and polarization textures

The skyrmion number Q of a polarization texture is defined by Eq. (1). For an optical field $\mathbf{E} = (E_x, E_y)$, the Stokes parameters are

$$S_0 = |E_x|^2 + |E_y|^2, \tag{2}$$

$$S_1 = |E_x|^2 - |E_y|^2, \tag{3}$$

$$S_2 = 2\,\mathrm{Re}(E_x E_y^*), \tag{4}$$

$$S_3 = 2\,\mathrm{Im}(E_x E_y^*), \tag{5}$$

and the normalized Stokes vector is $\mathbf{n} = (S_1, S_2, S_3)/S_0$.

A skyrmionic polarization field with charge q is constructed by clamping the polar angle $\theta = \pi \cdot \mathrm{clip}(r/R, 0, 1)$ to the disk of radius R, so that $\mathbf{n} \to (0, 0, -1)$ at $r \geq R$. The Jones field is

$$E_x = \cos(\theta/2), \tag{6}$$

$$E_y = \sin(\theta/2) e^{iq\phi}, \tag{7}$$

where $\phi = \mathrm{atan2}(y, x)$ is the azimuthal angle. This construction guarantees $Q = q$ exactly on any grid that resolves the texture.

2.2 Atmospheric turbulence model

We model atmospheric turbulence using the von Karman spectrum of refractive index fluctuations:

$$\Phi_n(\kappa) = 0.033\, C_n^2\, \kappa^{-11/3} \exp(-\kappa^2/\kappa_m^2), \tag{8}$$

where $\kappa_m = 5.92/l_0$ is the inner-scale cutoff, l_0 is the inner scale, and modes below the outer scale $\kappa_0 = 2\pi/L_0$ are zeroed. Each propagation slice of thickness dz generates independent phase and log-amplitude screens from this spectrum, scaled by dz so that the total turbulence is independent of the number of screens. The field is propagated using split-step Fourier alternation of free-space Fresnel propagation and screen application:

$$E \to E \exp(\chi + i\phi), \tag{9}$$

where χ is the log-amplitude screen and ϕ is the phase screen.

The phase screen is normalized to theoretical von Karman variance: $\sigma_\phi^2 = (2\pi/\lambda)^2 \int \Phi_n(\kappa)\, d^2\kappa$. The log-amplitude screen is normalized with

$\mathrm{Var}(\chi) = 0.189 \int \Phi_n(\kappa)\, d^2\kappa$, where $0.189 \approx 0.55/2.91$ is the Rytov-theory ratio of amplitude to phase variance [3].

Two turbulence regimes are distinguished:

- **Phase-only turbulence** (amplitude_turbulence=False): applies only phase screens, preserving polarization and total power exactly. This is an idealization useful for isolating phase effects.
- **Full turbulence** (amplitude_turbulence=True): applies both phase and log-amplitude screens, causing scintillation (intensity fluctuations) in addition to phase distortion. This is the physically realistic regime.

2.3 Fried parameter and Rytov variance

The Fried parameter r_0 characterizes the atmospheric coherence length:

$$r_0 = \left[0.423\, k^2 \int C_n^2(z)\, dz\right]^{-3/5}, \tag{10}$$

where $k = 2\pi/\lambda$. For constant C_n^2 over path length L: $r_0 = [0.423\, k^2\, C_n^2\, L]^{-3/5}$.

The Rytov variance $\sigma_R^2 = 1.23\, C_n^2\, k^{7/6}\, L^{11/6}$ distinguishes weak turbulence ($\sigma_R^2 < 1$) from strong turbulence ($\sigma_R^2 > 1$).

3 Methods

3.1 Simulation setup

All simulations use a square computational grid with spatial sampling $dx = dy$ and extent 6–8 m. The wavelength is $\lambda = 1550$ nm. Turbulence is modeled with 5–10 phase screens along the propagation path, using the von Karman spectrum with inner scale $l_0 = 5$ mm and outer scale $L_0 = 10$ m.

Three modulation schemes are compared under *identical* constraints:

1. **SKM**: Skyrmion number $Q \in \{-1, 1\}$ (2-state) or $Q \in \{-3, \ldots, 4\}$ (8-state) constellation, transmitted as polarization textures via Eq. (7).
2. **OAM**: Laguerre–Gaussian modes LG_0^l with $l \in \{-1, 1\}$ (2-mode) or $l \in \{-3, \ldots, 4\}$ (8-mode), detected by overlap with each mode field.
3. **Intensity**: Gaussian beam at power levels $\{0.5, 1.5\}$ (2-level) or $\{0.4, 0.7, \ldots, 2.5\}$ (8-level), detected by total integrated power.

All schemes transmit the same total optical power (normalized to 1.0 W integrated over the grid), use the same grid, wavelength, distance, screen count, and turbulence realization (shared phase and amplitude screens per Cn^2 value). Trials are statistically independent: fresh turbulence screens are generated for each trial.

3.2 Receiver model

The receiver measures the full Stokes field via polarimetry and computes the skyrmion number Q from the measured polarization texture using the gauge-invariant lattice solid-angle (plaquette) method:

$$Q = \frac{1}{4\pi} \sum_{\text{plaquettes}} \Omega, \tag{11}$$

where Ω is the solid angle subtended by the four normalized Stokes vectors at each plaquette. The received symbol is the nearest constellation value to the measured Q.

Receiver noise is modeled as additive complex Gaussian detector noise with standard deviation relative to the RMS field amplitude, and optionally as Poisson shot noise with mean photoelectrons per pixel.

3.3 BER and mutual information

Bit error rate (BER) is the fraction of trials with at least one bit error. Symbol error rate (SER) equals BER for single-symbol transmission. Mutual information $I(X;Y)$ is computed from the confusion matrix using $I(X;Y) = \sum_{x,y} P(x,y) \log_2[P(x,y)/(P(x)P(y))]$. Clopper–Pearson exact 95% confidence intervals are reported for all BER measurements.

4 Results

4.1 Turbulence model validation

The turbulence phase screen variance matches the von Karman spectrum integral to machine precision (relative error $< 10^{-12}$). Total turbulence is independent of the number of screens (phase standard deviation spread < 0.1 rad across $n_{\text{screens}} \in \{1, 5, 10, 20\}$). Split-step propagation conserves power exactly in the phase-only regime (relative error $\sim 10^{-15}$) and preserves mean power with scintillation (relative error $\sim 10^{-6}$).

The Fried parameter r_0 matches the theoretical formula $[0.423\,k^2\,C_n^2\,L]^{-3/5}$ to machine precision at all tested C_n^2 values from 10^{-17} to 10^{-12} m$^{-2/3}$ (Table 1).

Table 1: Fried parameter validation. r_0 computed from the turbulence model matches theory to machine precision at all C_n^2 values. $\lambda = 1550$ nm, $L = 1000$ m.

C_n^2 (m$^{-2/3}$)	r_0 computed (m)	r_0 theory (m)	Regime
10^{-17}	4.9520	4.9520	Weak
10^{-16}	1.2439	1.2439	Weak
10^{-15}	0.3124	0.3124	Weak
10^{-14}	0.0785	0.0785	Weak/Strong
10^{-13}	0.0197	0.0197	Strong
10^{-12}	0.0050	0.0050	Strong

4.2 Skyrmion number quantization

The skyrmion number Q is exactly integer for all tested charges $q \in \{-4, -3, -2, -1, 1, 2, 3, 4\}$ (Table 2). The solid-angle plaquette method returns $Q = q$ with relative error $< 2.3 \times 10^{-16}$ (machine precision). Grid independence is confirmed: $Q = 2$ exactly at grid resolutions 32×32 through 512×512.

Table 2: Skyrmion number quantization. $Q = q$ exactly for all tested charges at 128×128 grid, extent 8 m, radius 1 m.

Charge q	Q computed	Round(Q)	Exact?
-4	-4.000000	-4	Yes
-3	-3.000000	-3	Yes
-2	-2.000000	-2	Yes
-1	-1.000000	-1	Yes
1	1.000000	1	Yes
2	2.000000	2	Yes
3	3.000000	3	Yes
4	4.000000	4	Yes

4.3 Bit error rate comparison

2-state constellation, $Cn^2 = 10^{-13}$, 100 m link, ideal receiver: SKM achieves BER = 0.0000 (0/50 trials), OAM achieves BER = 0.5000 (25/50 trials), and intensity modulation achieves BER = 0.0000 (0/50 trials). SKM's advantage over OAM is infinite (BER = 0 vs 0.5).

Table 3 summarizes BER across turbulence strengths. SKM maintains BER = 0 at all tested turbulence levels ($C_n^2 \leq 10^{-11}$), while OAM is destroyed (BER ≈ 0.5) at all turbulence levels. Intensity modulation ties with SKM at low turbulence because total power is conserved by the split-step propagator even with scintillation; the 3× level separation (0.5 vs 1.5) is far larger than turbulence-induced power fluctuations.

Table 3: BER comparison under identical constraints. 50 trials per configuration, ideal receiver, 2-state constellation, 100 m link, 5 turbulence screens.

C_n^2	SKM BER	OAM BER	Intensity BER	SKM advantage
10^{-15}	0.0000	0.0000	0.0000	tie
10^{-14}	0.0000	0.1600	0.0000	infinite
10^{-13}	0.0000	0.5000	0.0000	infinite
10^{-12}	0.0000	0.5200	0.0000	infinite
10^{-11}	0.0000	0.5000	0.0000	infinite

Receiver noise sensitivity: At receiver noise standard deviation ≤ 0.2 (relative to RMS field amplitude), SKM maintains BER = 0 while OAM has BER ≈ 0.46. At noise 0.3, SKM degrades to BER = 0.28 while intensity remains at BER = 0. Simple power detection is more robust to additive shot noise than topological charge measurement.

High-trial validation: At $C_n^2 = 10^{-13}$ with 5,000 trials, SKM achieves BER = 0 with Clopper–Pearson 95% CI upper bound $< 6 \times 10^{-4}$. OAM achieves BER ≈ 0.5.

4.4 8-state constellation comparison

At $C_n^2 = 10^{-13}$ with 3 bits per symbol, SKM achieves BER = 0.0000 (0/100 trials) with mutual information 2.9634/3 bits. OAM achieves BER = 0.6900 (69/100 trials) with MI 2.4194/3 bits. Intensity achieves BER = 0.6300 (63/100 trials) with MI 1.6107/3 bits. SKM's topological protection scales to higher-order constellations.

4.5 Distance scaling

At fixed $C_n^2 = 10^{-13}$, SKM maintains BER = 0 at distances from 10 m to 500 m. OAM BER varies unpredictably (0, 1, 1, 0, 0.5) as turbulence destroys OAM mode structure. Intensity remains BER = 0 at all distances due to power conservation.

5 Discussion

5.1 Interpretation of results

SKM's advantage over OAM is decisive and consistent: topological protection preserves the skyrmion number under turbulence that scrambles OAM phase structure. This advantage holds across turbulence strengths, distances, and constellation orders.

The comparison with intensity modulation is more nuanced. Intensity modulation ties with SKM at low receiver noise because total optical power is conserved by the unitary split-step propagator. At high receiver noise, simple power detection outperforms topological charge measurement because it avoids the nonlinear demodulation step. SKM regains the advantage in higher-order constellations where intensity levels become difficult to distinguish.

5.2 Limitations

All results in this paper are from mathematical simulation on a laptop computer. The following require physical laboratory equipment:

1. **Physical FSO link**: A laser source, polarization modulator, free-space path, turbulence chamber (or outdoor link), and polarization-resolving receiver are needed to demonstrate SKM experimentally.

2. **Real photon statistics**: Our shot noise model assumes Poisson statistics; real detectors have dark counts, afterpulsing, and dead time not captured here.

3. **Calibration errors**: Real polarimeters have calibration errors, cross-talk between Stokes parameters, and finite extinction ratio that are not modeled.

4. **Finite aperture effects**: Our simulations use infinite-aperture detection (integration over the full grid). Real receivers have finite apertures that clip the beam, causing power fluctuations not captured here.

5.3 Levels of success

We distinguish three levels of success:

1. **Theoretical validity**: The skyrmion number is a topological invariant, and our Jones field construction guarantees $Q = q$ exactly. This is established analytically and verified numerically to machine precision.

2. **Computational demonstration**: Our simulation framework demonstrates SKM's BER advantage over OAM under identical constraints, validated against von Karman spectrum theory, Fried parameter formulae, and Rytov variance. This is the level achieved in this paper.

3. **Physical experimental demonstration**: Requires laboratory hardware not available to us. This remains future work.

We never confuse these levels. Computational results do not establish physical feasibility.

6 Conclusion

Skyrmion number modulation is a topologically protected alternative to OAM and intensity modulation for free-space optical communication. Under identical constraints, SKM achieves BER $= 0$ where OAM suffers BER ≈ 0.5, across turbulence strengths from weak to strong and across constellation orders from 2 to 8 states. The simulation framework is validated against von Karman spectrum theory, Fried parameter formulae, and Rytov variance, with skyrmion number quantization confirmed to machine precision at all grid resolutions.

Physical hardware demonstration is required to establish experimental feasibility. All source code, simulation parameters, and raw results are available for independent verification.

Acknowledgments

This research is independent and unaffiliated with any institution. All simulations are mathematical models on a laptop computer.

References

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